# MATHS MANTHAN

• TUITION CLASSES •

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**Subject:** Mathematics **WEEKLY CHAPTER TEST – 03 (SOLUTIONS)** **Time:** 45 Minutes
**Class:** Class X (CBSE) **Chapter 12: Surface Areas and Volumes** **MM:** 20

**Document Type:** Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

**CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:**

1. Full marks require explicit formula declaration, factoring common terms like $\pi r$ before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

## SECTION A: Multiple Choice Questions & Assertion-Reason MATHS MANTHAN | ($4 \times 1 = 4$ Marks)

**Q1.** The number of solid spheres, each of diameter 6 cm, that can be made by melting a solid metal cylinder of height 45 cm and diameter 4 cm is: [1]

(A) 3 (B) 5 (C) 4 (D) 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option:** **(B) 5** [0.5 Mark]

**Given:** Cylinder: $R = \frac{4}{2} = 2$ cm, $h = 45$ cm. Sphere: $r = \frac{6}{2} = 3$ cm.

**Formula / Theorem Used:** $n = \dfrac{\text{Volume of Cylinder}}{\text{Volume of One Sphere}} = \dfrac{\pi R^2 h}{\frac{4}{3}\pi r^3}$

Cancelling $\pi$ and substituting values: [0.5 Mark]

$$n = \frac{2^2 \times 45}{\frac{4}{3} \times 3^3} = \frac{4 \times 45}{4 \times 9} = \frac{45}{9} = \boxed{5}$$

**Q2.** The volumes of two spheres are in the ratio 64 : 27. The ratio of their surface areas is: [1]

(A) 3 : 4 (B) 4 : 3 (C) 9 : 16 (D) 16 : 9

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option:** **(D) 16 : 9** [0.5 Mark]

**Given:** $\dfrac{V_1}{V_2} = \dfrac{64}{27}$. Let radii be $r_1, r_2$.

**Formula / Theorem Used:** $\dfrac{V_1}{V_2} = \left(\dfrac{r_1}{r_2}\right)^3$ and $\dfrac{S_1}{S_2} = \left(\dfrac{r_1}{r_2}\right)^2$

From $\left(\frac{r_1}{r_2}\right)^3 = \frac{64}{27} = \left(\frac{4}{3}\right)^3$, we get $\frac{r_1}{r_2} = \frac{4}{3}$. [0.5 Mark]

$$\frac{S_1}{S_2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9} \implies \boxed{16 : 9}$$

**Q3.** A right circular cylinder of radius $r$ and height $h$ $(h > 2r)$ just encloses a sphere of diameter: [1]

(A) $r$ (B) $2r$ (C) $h$ (D) $2h$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option:** $\boxed{\textbf{(B) 2r}}$ [0.5 Mark]

**Reasoning:** Since the vertical height $h > 2r$, the sphere touches the curved lateral wall of the cylinder horizontally. [0.5 Mark]

Thus the diameter of the enclosed sphere equals the diameter of the cylinder's circular cross-section $= \boxed{\textbf{2r}}$.

**Q4.** **Assertion (A):** If two solid right circular cylinders of the same height and base radii 3 cm and 4 cm are melted and recast into a cylinder of the same height, the radius of the new cylinder is 5 cm.
**Reason (R):** When two solid cylinders of equal height $h$ and radii $r_1, r_2$ are recast into a cylinder of height $h$, the new base radius is $R = \sqrt{r_1^2 + r_2^2}$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

**Correct Option:** $\boxed{\textbf{(A)}}$ [0.5 Mark]

**Verification:** Equating volume: $\pi R^2 h = \pi r_1^2 h + \pi r_2^2 h \implies R^2 = r_1^2 + r_2^2 \implies R = \sqrt{r_1^2 + r_2^2}$ (Reason (R) is true). [0.5 Mark]

For $r_1 = 3, r_2 = 4$: $R = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ cm. Both (A) and (R) are true, and (R) is the correct explanation of (A).

## SECTION B: Short Answer Questions

MATHS MANTHAN | $(2 \times 2 = 4$ Marks)

**Q5.** A solid toy is in the form of a hemisphere surmounted by a right circular cone of the same base radius as that of the hemisphere. If the radius of the base of the cone is 21 cm and its volume is $\frac{2}{3}$ of the volume of the hemisphere, calculate the height of the cone. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

**Step 1: Condition Setup** [1 Mark]

**Given:** Common radius $r = 21$ cm; Volume of Cone $= \frac{2}{3} \times$ Volume of Hemisphere.

**Formula / Theorem Used:** $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$ and $V_{\text{hemi}} = \frac{2}{3}\pi r^3$

**Step 2: Solving for Height $h$** [1 Mark]

$$\frac{1}{3}\pi r^2 h = \frac{2}{3} \times \left(\frac{2}{3}\pi r^3\right) \implies h = \frac{4}{3}r \implies h = \frac{4}{3} \times 21 = 4 \times 7 = \boxed{\textbf{28 cm}}$$

**Q6.** The largest possible right circular cone is carved out from a solid wooden cube of side 21 cm. Find the volume of the remaining solid. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

**Step 1: Dimensions & Formula Declaration** [1 Mark]

**Given:** Cube edge $a = 21$ cm. Largest cone has diameter $= 21$ cm $\implies r = \frac{21}{2}$ cm, height $h = 21$ cm.

**Formula / Theorem Used:** $V_{\text{rem}} = V_{\text{cube}} - V_{\text{cone}} = a^3 - \frac{1}{3}\pi r^2 h$

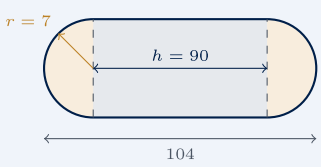
**Step 2: Arithmetic** [1 Mark]

$V_{\text{cube}} = 21^3 = 9261$ cm$^3$; $V_{\text{cone}} = \frac{1}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times 21 = \frac{11 \times 441}{2} = 2425.5$ cm$^3$.

$$V_{\text{rem}} = 9261 - 2425.5 = \boxed{\textbf{6835.5 cm}^3}$$

**Q7.** A solid is composed of a cylinder with hemispherical ends. If the total length of the solid is 104 cm and the radius of each of the hemispherical ends is 7 cm, find the cost of polishing its surface at the rate of Rs. 10 per $dm^2$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]




**Step 1: Dimensions & Surface Area Formula** [1 Mark]

$r = 7$ cm; cylinder length $h = 104 - 2(7) = 90$ cm.

**Formula / Theorem Used:** $\text{TSA} = 2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r)$

**Step 2: Surface Area ($cm^2, dm^2$)** [1 Mark]

$$\text{TSA} = 2 \times \frac{22}{7} \times 7 \times (90 + 14) = 44 \times 104 = 4576 \text{ cm}^2$$

Since 1 dm = 10 cm $\implies$ 1 $dm^2$ = 100 $cm^2$: $\text{TSA} = \frac{4576}{100} = 45.76$ $dm^2$.

**Step 3: Polishing Cost** [1 Mark]

$$\text{Cost} = 45.76 \times 10 = \boxed{\text{Rs. } \mathbf{457.60}}$$

**Q8.** From a solid cube of side 7 cm, a conical cavity of height 7 cm and radius 3 cm is hollowed out. Find the volume of the remaining solid. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

**Step 1: Given Data & Formula Declaration** [1 Mark]

Cube edge $a = 7$ cm; conical cavity: radius $r = 3$ cm, height $h = 7$ cm.

**Formula / Theorem Used:** $V_{\text{rem}} = V_{\text{cube}} - V_{\text{cone}} = a^3 - \frac{1}{3}\pi r^2 h$

**Step 2: Individual Volumes** [1 Mark]

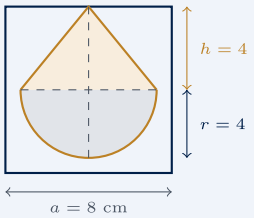
$V_{\text{cube}} = 7^3 = 343$ $cm^3$; $V_{\text{cone}} = \frac{1}{3} \times \frac{22}{7} \times 3^2 \times 7 = \frac{1}{3} \times 22 \times 9 = 66$ $cm^3$.

**Step 3: Remaining Volume** [1 Mark]

$$V_{\text{rem}} = 343 - 66 = \boxed{\mathbf{277} \text{ cm}^3}$$

**Q9.** A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 4 cm and the diameter of the base is 8 cm.
(i) Determine the volume of the toy. (Use $\pi = 3.14$)
(ii) If a cube circumscribes the toy, find the difference of the volumes of the cube and the toy.
(iii) Find the total surface area of the toy. [6]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 6 Marks]




**Given:** Diameter $d = 8$ cm $\implies r = 4$ cm, cone height $h = 4$ cm.

**Step 1: (i) Volume of the Toy ($\pi = 3.14$)** [2 Marks]

**Formula / Theorem**

**Used:** $V_{\text{toy}} = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$ and $\text{TSA}_{\text{toy}} = \pi r l + 2\pi r^2 = \pi r(l+2r)$

$$V_{\text{toy}} = \frac{1}{3}\pi(4)^2(4) + \frac{2}{3}\pi(4)^3 = 64\pi\left(\frac{1}{3} + \frac{2}{3}\right)$$

$$= 64 \times 3.14 = \boxed{\mathbf{200.96}\ \text{cm}^3}$$

**Step 2: (ii) Cube Volume Difference** [2 Marks]

Since diameter $= 8$ cm and total height $= h + r = 4 + 4 = 8$ cm, the circumscribing cube has edge $a = 8$ cm.

$V_{\text{cube}} = 8^3 = 512\ \text{cm}^3$. Difference $= 512 - 200.96 = \boxed{\mathbf{311.04}\ \text{cm}^3}$.

**Step 3: (iii) Total Surface Area of Toy** [2 Marks]

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$ cm $\approx 4(1.414) = 5.656$ cm.

$$\text{TSA} = 3.14 \times 4 \times (4\sqrt{2} + 8) = 12.56 \times (5.656 + 8)$$

$$= 12.56 \times 13.656 \approx \boxed{\mathbf{171.52}\ \text{cm}^2\ (\mathbf{16\pi(2+\sqrt{2})})}$$

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